Project Title - Seasonal Agriculture Performance Analysis

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PROBLEM STATEMENT

- (Agricultural activities are influenced by seasonal variations in environmental conditions, farming practices, resource availability and market conditions. As a result, agricultural performance may differ from one season to another. However, raw agricultural data does not clearly explain how agricultural performance changes across seasons or what patterns can be observed in different seasonal conditions. The problem is to analyze the given agricultural dataset and investigate seasonal differences in agricultural performance by identifying meaningful patterns, trends, relationships and variations within the available data.

Project Description

The Seasonal Agriculture Performance Analysis project focuses on analyzing agricultural data collected across different seasons, crops and farming conditions.

The project begins with understanding and cleaning the agricultural dataset by checking its structure, data types, missing values and duplicate records. Exploratory Data Analysis is then performed to understand the characteristics of the data.

The main focus of the project is to compare agricultural performance across different seasons. Seasonal analysis is performed using relevant agricultural measures such as yield, production, rainfall, temperature, resource usage and economic factors, depending on the variables available in the dataset.

Crop-wise and crop-season comparisons are also performed to identify differences in agricultural performance. Relationships between environmental conditions and agricultural outcomes are explored using correlation analysis and suitable visualizations.

Graphs and charts are used to clearly present seasonal patterns, trends and variations. The findings are interpreted based on the actual data, and data-driven recommendations are provided to support better seasonal agricultural planning.

WHO ARE THE END USERS?

- **Farmers** – to understand seasonal crop performance and resource usage.
- **Agricultural planners** – to support seasonal agricultural planning.
- **Agricultural organizations** – to understand patterns in agricultural production.
- **Researchers and students** – to study seasonal agricultural trends.
- **Government and agricultural departments** – to support evidence-based agricultural decisions.
- **Data analysts** – to analyze agricultural performance and identify useful patterns.

Technology Used

- **Python**
- **Google Colab / Jupyter Notebook**
- **Pandas** – Data cleaning and analysis
- **NumPy** – Numerical operations
- **Matplotlib** – Data visualization
- **Seaborn** – Statistical visualization
- **Excel** – Dataset
- **GitHub** – Project documentation and code sharing

RESULTS

Seasonal Agricultural Performance Analysis.ipynb

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[51]

```
## – Dataset Overview

# Load the dataset into a pandas DataFrame named 'df'
# Assuming your dataset is in '/content/seasonal_agriculture_performance_dataset.csv'
# If your file has a different name or path, please update it here.
df = pd.read_csv('/content/seasonal_agriculture_performance_dataset.csv')

print("--- Dataset Shape ---")
print(df.shape)

print("\n--- First 5 Rows ---")
display(df.head())

print("\n--- Column Names and Data Types ---")
df.info()

print("\n--- Exact Column Names List ---")
print(df.columns.tolist())
```

```
--- Dataset Shape ---
(4000, 28)

--- First 5 Rows ---
```

	Farm_ID	State	District	Crop	Season	Farm_Area_Hectares	Rainfall_mm	Avg_Temperature_C	Humidity_pct	Sunlight_Hours_Day	...	Seed_Quality_Score	Yield_Tonnes_Ha	Production_Tonnes	Market_Price_INR_Tonne	Total_Cost_INR	Revenue_INR	Profit_INR	Water_Used_m3
0	SF10001	Andhra Pradesh	Rajkot	Wheat	Kharif	0.53	486.4	24.3	69.1	5.8	...	0.76	2.46	1.30	23700	42662	30810	-11852	237
1	SF10002	Maharashtra	Nalgonda	Maize	Kharif	6.53	855.4	25.7	78.4	5.1	...	0.94	0.30	1.96	20613	492351	40401	-451950	4953
2	SF10003	Telangana	Warangal	Pulses	Rabi	4.86	455.6	23.9	56.4	10.3	...	0.98	0.52	2.53	75283	315210	190466	-124744	1275
3	SF10004	Telangana	Indore	Rice	Kharif	4.50	753.2	29.7	65.2	6.3	...	0.74	1.84	8.28	24244	296444	200740	-95704	3834
4	SF10005	Karnataka	Ludhiana	Maize	Zaid	4.21	101.6	34.1	55.2	6.5	...	0.90	2.21	9.30	20818	337959	193607	-144352	3287

5 rows × 28 columns

```
--- Column Names and Data Types ---
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 4000 entries, 0 to 3999
Data columns (total 28 columns):
 #   Column                  Non-Null Count  Dtype
---  ------                  --------------  -----
 0   Farm_ID                 4000 non-null   object
 1   State                   4000 non-null   object
 2   District                4000 non-null   object
 3   Crop                    4000 non-null   object
 4   Season                  4000 non-null   object
 5   Farm_Area_Hectares      4000 non-null   float64
 6   Rainfall_mm             3952 non-null   float64
 7   Avg_Temperature_C       4000 non-null   float64
 8   Humidity_pct            4000 non-null   float64
 9   Sunlight_Hours_Day      4000 non-null   float64
 10  Soil_pH                 4000 non-null   float64
 11  Soil_Moisture_pct       3960 non-null   float64
```

RESULTS

CODE

```
## — Data Quality Check

print("--- Missing Values Check ---")
print(df.isnull().sum())

print("\n--- Duplicate Rows Check ---")
num_duplicates = df.duplicated().sum()
print(f"Number of duplicate rows: {num_duplicates}")

print("\n--- Basic Data Types (before detailed cleaning) ---")
df.info()
```

```
Soil_Moisture_pct              40
Nitrogen_kg_ha                  0
Phosphorus_kg_ha                0
Potassium_kg_ha                 0
Irrigation_Method               0
Fertilizer_kg_ha                0
Pesticide_Litre_ha              0
Seed_Quality_Score              0
Yield_Tonnes_Ha                32
Production_Tonnes               0
Market_Price_INR_Tonne          0
Total_Cost_INR                  0
Revenue_INR                     0
Profit_INR                      0
Water_Used_m3                   0
Water_Efficiency_t_per_1000m3   0
Disease_Pest_Risk_pct           0
dtype: int64

--- Duplicate Rows Check ---
Number of duplicate rows: 0
```

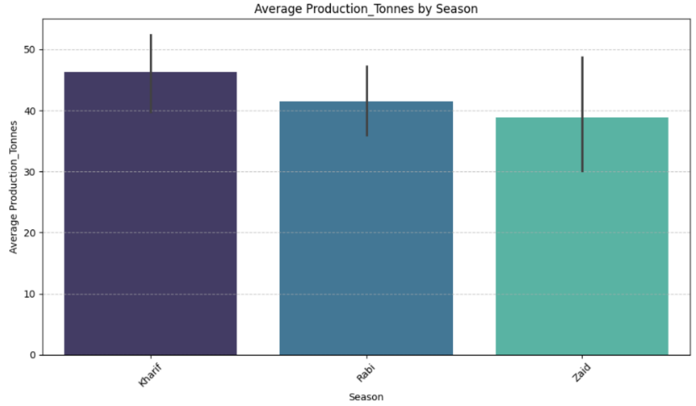
RESULTS

Season

```
/tmp/ipykernel_3848/3779580603.py:34: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same effect.

  sns.barplot(x=season_col, y=production_col, data=df, estimator=np.mean, palette='mako')
```




```
print("Season column not found.")
```

```
--- Seasonal Analysis ---
Using season column: Season
```

	Season	Number of Records	Average Yield	Average Production	Average Rainfall	Average Temperature	Average Fertilizer	Average Pesticide	Average Cost	Average Price	Average Profit
0	Kharif	1779	5.628612	46.311192	849.199663	28.454019	187.077965	5.079747	531804.408094	44850.906127	178914.647555
1	Rabi	1627	5.039435	41.486712	437.615366	23.489183	185.407068	5.024690	513836.578365	44747.974800	87689.470191
2	Zaid	594	4.641313	38.886111	304.654714	31.042088	184.638047	5.171212	543976.728956	44212.033670	-24804.824916

RESULTS

```python
# 3. Unusual rainfall (if rainfall_col exists)
if rainfall_col and rainfall_col in df.columns:
    mean_rainfall = df[rainfall_col].mean()
    std_rainfall = df[rainfall_col].std()
    high_rainfall_threshold = mean_rainfall + (2 * std_rainfall)

    unusually_high_rainfall = df[df[rainfall_col] > high_rainfall_threshold].sort_values(by=rainfall_col, ascending=False)
    if not unusually_high_rainfall.empty:
        print(f"\n--- Instances of Unusually High {rainfall_col} (Above {high_rainfall_threshold:.2f}) ---")
        display(unusually_high_rainfall.head())
    else:
        print(f"\nNo unusually high {rainfall_col} observations (above {high_rainfall_threshold:.2f}) found.")
else:
    print(f"Warning: '{rainfall_col}' column not found or not assigned. Skipping unusual rainfall analysis.")
```

--- Unusual Patterns / Significant Observations ---

--- Instances of Unusually High Yield_Tonnes_Ha (Above 31.87) ---

	Farm_ID	State	District	Crop	Season	Farm_Area_Hectares	Rainfall_mm	Avg_Temperature_C	Humidity_pct	Sunlight_Hours_Day	...	Seed_Quality_Score	Yield_Tonnes_Ha	Production_Tonnes	Market_Price_INR_Tonne	Total_Cost_INR	Revenue_INR	Profit_INR
1715	SF11716	Gujarat	Indore	Sugarcane	Kharif	6.32	689.0	29.6	68.2	5.9	...	0.75	101.44	641.10	3077	364374	1972665	1608291
3423	SF13424	Andhra Pradesh	Indore	Sugarcane	Kharif	14.87	745.9	29.6	74.1	6.4	...	0.82	95.79	1424.40	3109	1097878	4428460	3330582
3525	SF13526	Andhra Pradesh	Erode	Sugarcane	Rabi	6.96	772.5	27.9	64.5	7.2	...	0.73	93.13	648.18	2917	419212	1890741	1471529
1464	SF11465	Gujarat	Nashik	Sugarcane	Kharif	7.65	579.6	28.0	75.0	5.5	...	1.00	91.98	703.65	3839	484604	2701312	2216708
3523	SF13524	Karnataka	Nashik	Sugarcane	Kharif	0.82	752.4	29.5	67.5	5.8	...	0.85	91.46	75.00	3898	58030	292350	234320

5 rows × 28 columns

No unusually low Yield_Tonnes_Ha observations (below -21.38) found.
Cannot assess large seasonal differences due to missing seasonal summary or yield column.

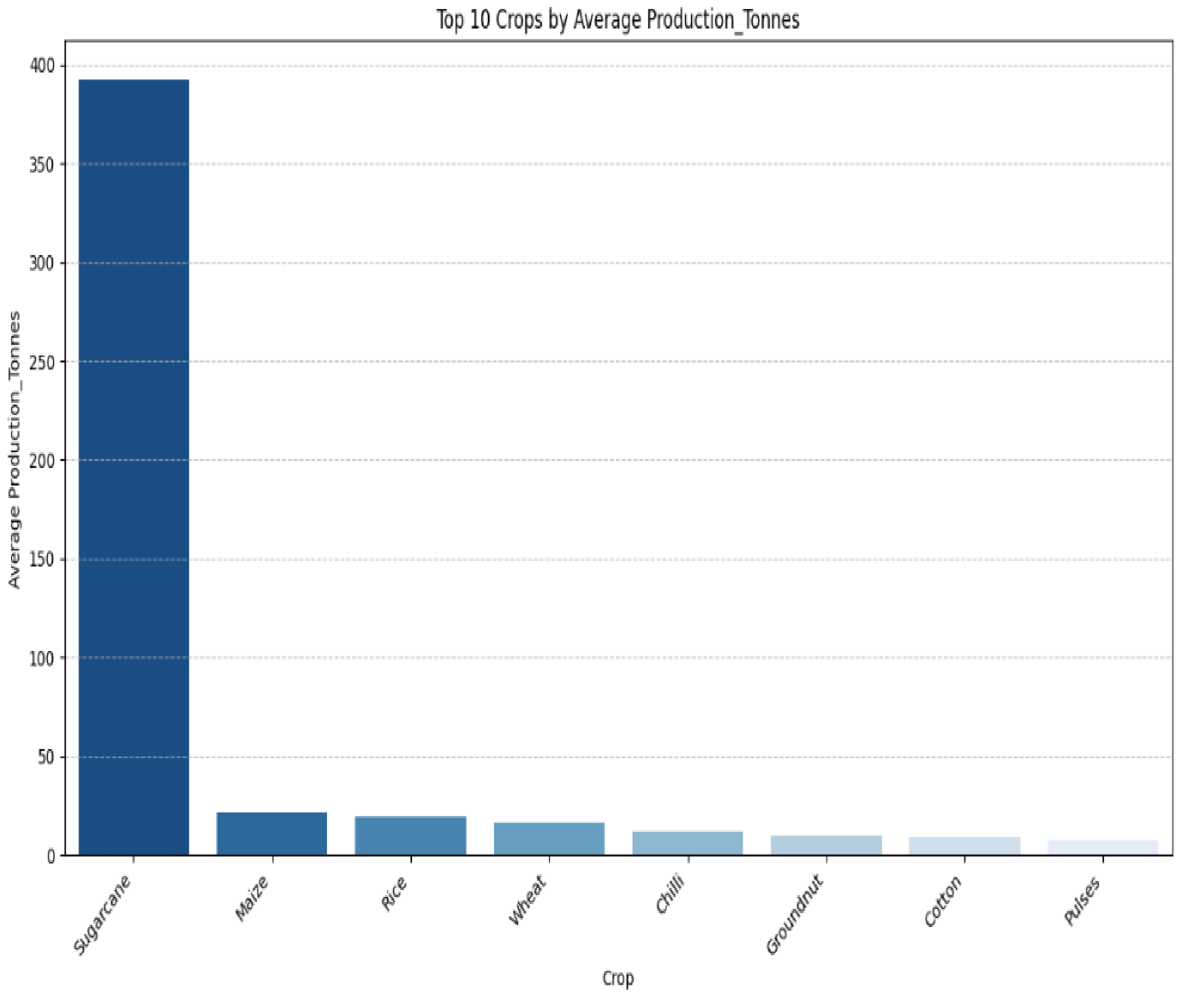
--- Instances of Unusually High Rainfall_mm (Above 1175.31) ---

	Farm_ID	State	District	Crop	Season	Farm_Area_Hectares	Rainfall_mm	Avg_Temperature_C	Humidity_pct	Sunlight_Hours_Day	...	Seed_Quality_Score	Yield_Tonnes_Ha	Production_Tonnes	Market_Price_INR_Tonne	Total_Cost_INR	Revenue_INR	Profit_INR
1495	SF11496	Telangana	Guntur	Wheat	Kharif	3.86	1395.2	25.9	64.4	6.2	...	0.87	2.40	9.26	22646	231318	209702	-21616
2887	SF12888	Karnataka	Raichur	Chilli	Kharif	3.46	1367.2	29.1	83.6	5.3	...	0.80	0.83	2.87	115645	224158	331901	107743
2247	SF12248	Karnataka	Rajkot	Wheat	Kharif	12.22	1356.9	33.7	82.5	5.8	...	0.89	1.26	15.40	22754	1057159	350412	-706747
791	SF10792	Tamil Nadu	Erode	Sugarcane	Kharif	13.49	1353.1	28.2	63.7	8.2	...	0.82	44.78	604.08	3909	1039127	2361349	1322222
2435	SF12436	Tamil Nadu	Indore	Maize	Kharif	14.15	1338.9	32.8	80.8	6.8	...	0.91	3.01	42.59	22048	975259	939024	-36235

5 rows × 28 columns

RESULTS

```
sns.barplot(x=avg_production_by_crop.head(10).index, y=avg_production_by_crop.head(10).values, palette='Blues_r')
```



Future scope

- Adding real-time agricultural and weather data.
- Including more geographical regions.
- Developing an interactive agricultural dashboard.
- Adding crop yield prediction using Machine Learning.
- Integrating real-time weather information.
- Developing a recommendation system for crop and season selection.
- Including larger datasets covering multiple years.
- Adding advanced economic and market analysis.

GitHub Link

https://github.com/saikirane030/Seasonal-Agriculture-Performance-Analysis

VOIS Course completion certificate - Course name- Data Visualization


Vodafone Idea Foundation

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Certificate of Completion

Presented to

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For the successful completion of

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COURSE COMPLETION

Thank you

https://github.com/saikirane030/Seasonal-Agriculture-Performance-Analysis